

# Natural Disaster Severity Prediction via Gradient Boosting and Temporal Feature Engineering

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**Abstract**—Group ID: 5. Members: Hsin-Yu Chen (314551141, sychen.cs14@nycu.edu.tw), Wei-Hsin Hung (314553021, velda.cs14@nycu.edu.tw), and Sky Shih-Kai Hong (314554024, sky.cs14@nycu.edu.tw). Our code is available at GitHub: <https://github.com/skyhong2002/disaster-severity-prediction>.

Natural disaster severity prediction is a critical task for risk management and early warning systems. In this work, we address the problem of forecasting weekly drought severity scores (0–5 scale) for 2,248 geographic regions using historical meteorological data. We employ a leakage-aware direct multi-step forecasting approach using an ensemble of LightGBM, XGBoost, and CatBoost models. The implemented system builds temporal features per horizon from calendar variables, meteorological lags, rolling statistics, exponentially weighted means, long-window drought proxies, climatology anomalies, and 91-day-gapped historical score context. By separating historical diagnostic experiments from the current direct-horizon implementation, we avoid accidental score leakage while preserving useful region-specific signals. The current best legal Kaggle public leaderboard score recorded by the repository is 0.8124.

## I. PROJECT SUMMARY (MAX 1 PAGE)

### A. Goal

The goal of this project is to predict the drought severity score for each geographic region over the **next five consecutive weeks** (35 days), given 91 days of historical meteorological observations per region from the test set. Scores range from 0 (no disaster) to 5 (extreme severity).

### B. High-Level Approach

Our approach follows a supervised machine learning paradigm treating the task as a **multi-step time series regression** problem. We extract temporal features from the raw meteorological columns, including lag values, rolling means/stds, exponentially weighted averages, calendar encodings, climatology anomalies, and domain-specific drought indices. We also include historical score features only after a 91-day gap, matching the blind test window and preventing access to unavailable labels. These features feed into direct horizon models using LightGBM, XGBoost, and CatBoost. To maximize robustness, we ensemble their predictions. Final predictions are post-processed by clipping values to the valid score interval [0, 5].

### C. Related Works

Time series forecasting for environmental phenomena has been widely studied. Gradient boosting methods, notably XGBoost [1] and LightGBM [2], have demonstrated strong performance on tabular time series data by capturing non-linear feature interactions without requiring extensive feature scaling. For forecasting competitions and meteorological applications, lag-window feature engineering is often effective because recent observations and seasonal patterns usually contain strong predictive signals [3]. Transformer-based models such as Temporal Fusion Transformer [4] and PatchTST [5] are possible future extensions, but the current repository implementation focuses on a fully reproducible gradient-boosted tree pipeline.

## II. PROPOSED METHOD (1-3 PAGES)

### A. Problem Formulation

Let  $\mathcal{R} = \{r_1, r_2, \dots, r_K\}$  denote the set of geographic regions. For each region  $r_k$ , we observe a time series of meteorological features  $\mathbf{X}_{r_k} \in \mathbb{R}^{T \times D}$  where  $T$  is the number of observed days and  $D$  is the number of meteorological variables. The target  $y_{r_k, t}$  is the drought severity score at week  $t$ , available only on a specific day per week (the rest are NaN).

Given the 91-day test window, the task is to predict:

$$\hat{y}_{r_k, t+1}, \hat{y}_{r_k, t+2}, \dots, \hat{y}_{r_k, t+5} \quad (1)$$

minimizing the Mean Absolute Error:

$$\text{MAE} = \frac{1}{K \cdot 5} \sum_{k=1}^K \sum_{w=1}^5 |\hat{y}_{r_k, w} - y_{r_k, w}| \quad (2)$$

### B. Data Preprocessing

1) *Handling NaN Scores and Blind-Gap Features*: Scores are observed only once per week, so most daily rows have missing target values. During training, only rows with real non-missing scores are used as supervised labels. Earlier diagnostic experiments attempted to reconstruct missing scores with a separate model, but that design created confusing train-test behavior and is not the current implementation path. The current pipeline keeps historical score context only when it is available before the 91-day blind gap. A lag of zero in the score-history feature group therefore means “last known score”

before the blind window,” not the current or recent test-period score.

2) *Feature Engineering*: For each meteorological column  $d$ , we construct the following feature groups:

- **Meteorological lags**: 7, 14, 21, 28, 35, 42, and 49 days.
- **Rolling statistics**: shifted rolling mean and standard deviation over 7, 14, 21, 28, and 35 days.
- **EWM features**: exponentially weighted means with spans 3, 7, and 14 days.
- **Calendar features**: month, quarter, day of month, approximate day of year, week of year, and cyclical sine/cosine encodings.
- **Domain Features**: We introduced a Drought Index (temperature divided by precipitation) and a Dryness Index (daily temperature range multiplied by max temperature) to explicitly capture conditions favorable to drought.
- **Climatology anomalies**: region-month normal weather and standardized deviations for precipitation, temperature, humidity, and related signals.
- **Blind-gap score history**: score lags and rolling summaries shifted by at least 91 days to prevent leakage from the unlabeled test window.
- **Region statistics**: historical mean, standard deviation, median, maximum, and minimum score per region.

### C. Model Architecture

1) *Direct Horizon Models*: We train one independent regressor per prediction horizon (week 1 through week 5), a technique known as **Direct Multi-Step Forecasting**. Each horizon model uses the full feature set and predicts exactly one target column, `target_w1` through `target_w5`.

2) *Ensemble Strategy*: To capture diverse patterns and reduce variance, we train **LightGBM**, **XGBoost**, and **CatBoost** models. LightGBM uses leaf-wise growth, XGBoost is configured with histogram-based depth-wise trees, and CatBoost uses native categorical handling for features such as region and calendar buckets. The current best legal public submission uses a convex blend with weights 35% LightGBM, 35% XGBoost, and 30% CatBoost.

### D. Inference and Submission Generation

At inference time, recent training rows and the 91-day test rows are concatenated and sorted by `region_id` and `date`. We rebuild the feature table using the feature options saved in the selected experiment run, select the last row of the 91-day test window for each region, and apply the five saved horizon models. The inference script can load LightGBM, XGBoost, or CatBoost model pickles from a run directory. The predictions are then blended and outputted to a timestamped CSV file under `submissions/`.

### E. Experiment Versioning

Because the implementation is expected to evolve, each run is saved under a timestamped directory `experiments/<run_id>/`. The directory contains `config.json` for hyperparameters and feature

settings, `metrics.json` for validation results, and `submission_metadata.json` for prediction statistics after inference. Large model files and generated submission CSVs are stored inside the same run directory but are ignored by git. This format allows LightGBM, XGBoost, CatBoost, ensembles, and future neural time series models to be appended to the report as comparable experiment rows. Some historical run identifiers still use the string `two_stage`; this is retained for artifact compatibility and should not be interpreted as a description of the current training flow.

## III. EXPERIMENTS (2-4 PAGES)

1) *Dataset and Experimental Setup*: The dataset contains meteorological measurements for multiple geographic regions. The training set spans **5,480 days** per region with weekly drought scores (0–5). The test set provides 91 days of observations per region without labels; we must predict 5 future weekly scores.

Table I summarizes key dataset statistics.

TABLE I  
DATASET STATISTICS

| Property             | Train      | Test    |
|----------------------|------------|---------|
| Days per region      | 5,480      | 91      |
| # Regions            | 2,248      | 2,248   |
| # Raw meteo features | 14         | 14      |
| # Rows               | 12,319,040 | 204,568 |
| Score range          | 0–5        | N/A     |
| Score NaN ratio      | 85.73%     | N/A     |

The implementation supports both a **Chronological Hold-out Split** and a **Rolling-Origin** validation mode. The holdout split is used for the original LightGBM/XGBoost anchor runs, while rolling-origin validation is used for the CatBoost robustness experiments. These validation scores should not be compared as if they came from the same protocol; Kaggle public and private leaderboard results remain the external check for generalization.

2) *Baseline Descriptions*: We compare against the official Kaggle baselines reported on the public leaderboard. Baseline 1, Baseline 2, and Baseline 3 provide reference MAE scores for evaluating whether our implemented pipeline improves over simple competition-provided methods.

3) *Experimental Results*: Table II records the main legal experiments and historical diagnostics used to evaluate historical score features, validation behavior, and model diversity.

TABLE II  
EXPERIMENT LOG & KAGGLE LEADERBOARD EVALUATION

| Method Strategy                    | Local MAE       | Public MAE     |
|------------------------------------|-----------------|----------------|
| Naive ffill Persistence (gap=0)    | 0.2321 (Leaked) | 1.0815         |
| Strategy B (Pure Weather Direct)   | 0.6770          | 0.8640         |
| Long-Term Weather (365 days)       | 0.4888          | 0.8604         |
| LGB/XGB Anchor (50/50)             | 0.6942 / 0.7150 | 0.8232         |
| CatBoost tail2737 (rolling-origin) | 0.2212          | component only |
| <b>LGB/XGB/CatBoost 35/35/30</b>   | N/A             | <b>0.8124</b>  |
| Initial Submission (Magic Leak)    | 0.2100 (Leaked) | 0.8094*        |

\*Discarded due to unintentional data leakage in feature engineering.

Table III reports the Kaggle public leaderboard snapshot after our submissions. The public leaderboard is calculated with approximately 40% of the test data. The current best legal Team 5 public MAE recorded in the repository is **0.8124**, achieved by adding CatBoost as a third ensemble member while keeping the leaky diagnostic submissions excluded from official claims. The corresponding submission file is `submissions/ensemble_20260516_lgb_xgb_cat2737_35_35_30.csv`.

TABLE III  
KAGGLE PUBLIC LEADERBOARD SNAPSHOT

| Team / Baseline                  | Public MAE    | Entries  |
|----------------------------------|---------------|----------|
| Baseline 3                       | 0.8056        | 1        |
| Team 20                          | 0.8062        | 12       |
| <b>Team 5 (v0 Magic Leak)*</b>   | <b>0.8094</b> | <b>1</b> |
| <b>Team 5 (LGB/XGB/CatBoost)</b> | <b>0.8124</b> | <b>1</b> |
| <b>Team 5 (LGB/XGB Anchor)</b>   | <b>0.8232</b> | <b>3</b> |
| Team 4                           | 0.8392        | 37       |
| Team 3                           | 0.8534        | 15       |
| Baseline 2                       | 0.8623        | 1        |
| <b>Team 5 (Strategy B)</b>       | <b>0.8640</b> | <b>4</b> |
| Baseline 1                       | 0.9117        | 1        |
| Team 10                          | 1.1791        | 2        |

\*Discarded from final evaluation due to data leakage.

The generated submission contains one row per region and five predicted scores. Table IV summarizes the prediction distribution for the current best legal public submission. The mean forecast gradually decreases from week 1 to week 5, while all predictions remain inside the valid score range after clipping.

TABLE IV  
PREDICTION STATISTICS FOR CURRENT BEST LEGAL SUBMISSION

| Column     | Mean  | Std.  | Min   | Max   |
|------------|-------|-------|-------|-------|
| pred_week1 | 0.955 | 0.819 | 0.000 | 3.470 |
| pred_week2 | 0.927 | 0.818 | 0.000 | 3.376 |
| pred_week3 | 0.880 | 0.824 | 0.000 | 3.457 |
| pred_week4 | 0.831 | 0.804 | 0.000 | 3.428 |
| pred_week5 | 0.817 | 0.827 | 0.000 | 3.583 |

4) *Ablation Study: The Kaggle Paradox & Data Leakage:* To systematically analyze the impact of historical score dependencies and identify potential data leakage, we conducted an extensive ablation study.

**1. The “Magic Leak” Phenomenon (v0):** Our initial submission achieved a surprisingly strong Public MAE of 0.8094. However, investigation revealed this was caused by a train-test discrepancy where the model inadvertently relied on recent scores during training, while the testing phase relied on a fixed forward-filled (`ffill`) score. This created an accidental hybrid persistence model. To maintain scientific rigor, we discarded this model despite its high score.

**2. The Naive Persistence Failure (1.0815):** We explicitly reproduced the pure `ffill` behavior by setting the score gap to 0. While this yielded an artificially low Local MAE of 0.2321 (due to looking at the current day’s true score), it failed spectacularly on Kaggle with an MAE of 1.0815. This proves

that drought is highly dynamic and cannot be modeled by simply freezing the last known state for 91 days.

**3. The Kaggle Paradox & Long-Term Weather (0.8604):** We initially hypothesized that removing the noisy historical scores entirely (Strategy B) would improve generalization. We expanded this by providing the model with up to 365 days of accumulated weather data, successfully dropping the Local MAE to an impressive 0.4888. However, the Kaggle Public MAE remained at 0.8604. This paradox demonstrates that historical scores, even when noisy, contain crucial unobserved confounders (e.g., local water infrastructure, soil characteristics) that pure meteorological features cannot capture.

**4. The CatBoost Diversity Gain (0.8124):** After establishing the legal LGB/XGB anchor, we added a time-aware CatBoost direct-horizon model using native categorical handling and rolling-origin validation. A 35/35/30 blend of LightGBM, XGBoost, and CatBoost improved the legal public MAE from 0.8232 to 0.8124. Because the CatBoost validation protocol differs from the original holdout protocol, we treat the public improvement as promising but still require private leaderboard evidence before over-tuning its weight.

#### IV. CONCLUSION

In this work, we implemented a robust time series prediction framework for forecasting drought severity levels. By using chronological and rolling-origin validation modes and by enforcing 91-day-gapped score-history features, we reduced leakage risk while retaining useful regional severity context. Our ablation study revealed a compelling “Kaggle Paradox”: pure long-term meteorological models perform strongly locally but fail to generalize to the leaderboard because they lack unobserved geographical and infrastructural confounders implicitly encoded in historical drought scores. We demonstrated that naive target persistence fails completely (MAE 1.0815), while the current direct-horizon ensemble of LightGBM, XGBoost, and CatBoost achieves a competitive legal Public MAE of 0.8124.

#### REFERENCES

- [1] T. Chen and C. Guestrin, “Xgboost: A scalable tree boosting system,” in *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, 2016, pp. 785–794.
- [2] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, “Lightgbm: A highly efficient gradient boosting decision tree,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 3146–3154.
- [3] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, “M5 accuracy competition: Results, findings, and conclusions,” *International Journal of Forecasting*, vol. 38, no. 4, 2022.
- [4] B. Lim, S. Ö. Arık, N. Loeff, and T. Pfister, “Temporal fusion transformers for interpretable multi-horizon time series forecasting,” *International Journal of Forecasting*, vol. 37, no. 4, pp. 1748–1764, 2021.
- [5] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, “A time series is worth 64 words: Long-term forecasting with transformers,” in *International Conference on Learning Representations*, 2023.